Model selection

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Texas Suffers From A Doctor Shortage

By JONATHAN BAKER • NOV 1, 2017

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When it comes to having a high ratio of doctors to citizens, the State of Texas ranks near the bottom. In fact, as *The Dallas Morning News* reports, 43 states have a higher proportion of primary care physicians to residents than Texas.

And West Texas suffers from a lack of doctors more than other parts of the state. There are 80 counties in Texas with five or fewer practicing doctors - many in West Texas. Thirty-five Texas counties have no doctors at all.

The Texas counties data set

- All 168 Texas counties with 10,000+ residents.
- The response: **physicians per 10,000 people**, which runs from 0.0 to 12.9 across counties.
- Ten candidate predictors describing each county's land, people, and economy.

Our question: which of these factors predict how many doctors a county has?

Ten candidate predictors

`LandArea`	area in square miles	`PctOver65`	percent seniors
`PctRural`	percent rural land	`PctPoverty`	percent in poverty
`MedianIncome`	median household income	`PctUninsured`	percent uninsured
`Population`	population	`PctSomeCollege`	percent with some college
`PctUnder18`	percent children	`PctUnemployed`	percent unemployed

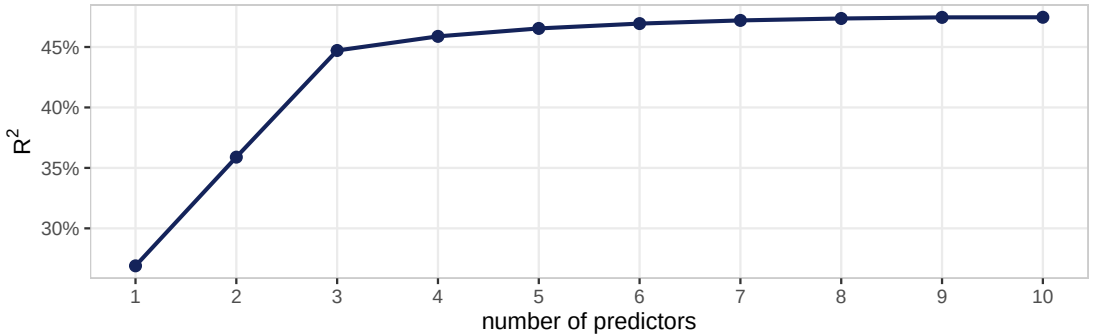
Which would you expect to matter? Which would you leave out?

With ten predictors there are 1,023 possible regressions.

How do we know which model is best?

Why not just use R^2?

The best model of each size, by R^2:



R^2 **never goes down** when you add a predictor. Maximize it and you always keep everything.

Suppose we want to predict the Democratic candidate's share of the vote in a presidential election...

THE PROBLEM WITH STATEMENTS LIKE

"NO <PARTY> CANDIDATE HAS WON THE ELECTION WITHOUT <STATE>"

OR

"NO PRESIDENT HAS BEEN REELECTED UNDER <CIRCUMSTANCES>"

★ UPDATED FOR 2020 ★

1788...
NO ONE HAS BEEN ELECTED PRESIDENT BEFORE.

...BUT WASHINGTON WAS.

1792...
NO INCUMBENT HAS EVER BEEN REELECTED.

...UNTIL WASHINGTON.

1796...
NO ONE WITHOUT FALSE TEETH HAS BECOME PRESIDENT.

...BUT ADAMS DID.

1800...
NO CHALLENGER HAS BEATEN AN INCUMBENT.

...BUT JEFFERSON DID.

1804...
NO INCUMBENT HAS BEATEN A CHALLENGER.

...UNTIL JEFFERSON.

1808...
NO CONGRESSMAN HAS EVER BECOME PRESIDENT.

...UNTIL MADISON.

1812...
NO ONE CAN WIN WITHOUT NEW YORK.

...BUT MADISON DID.

1816...
NO CANDIDATE WHO DOESN'T WEAR A WIG CAN GET ELECTED.

...UNTIL MONROE WAS.

1820...
NO ONE WHO WEARS PANTS INSTEAD OF BREECHES CAN BE REELECTED.

...BUT MONROE WAS.

1824...
NO ONE HAS EVER WON WITHOUT A POPULAR MAJORITY.

...J.Q. ADAMS DID.

1828...
ONLY PEOPLE FROM MASSACHUSETTS AND VIRGINIA CAN WIN.

...UNTIL JACKSON DID.

1832...
THE ONLY PRESIDENTS WHO GET REELECTED ARE VIRGINIANS.

...UNTIL JACKSON.

1836...
NEW YORKERS ALWAYS LOSE.

...UNTIL VAN BUREN.

1840...
NO ONE OVER 65 HAS WON THE PRESIDENCY.

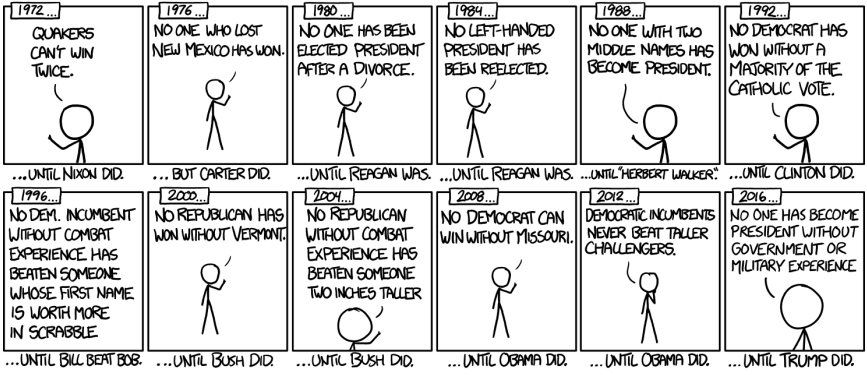
...UNTIL HARRISON DID.

1844...
NO ONE WHO'S LOST HIS HOME STATE HAS WON.

...BUT POLK DID.

1848...
AS GOES MISSISSIPPI, SO GOES THE NATION.

...UNTIL 1848.



CONGRATULATIONS TO PRESIDENT-ELECT JOE BIDEN FOR BREAKING THE WEBSITE CURSE!

Every one of these raises R^2

- Whether the candidate has won Missouri

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- The height difference between the candidates

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- The height difference between the candidates
- Whether the candidate has combat or government experience
- Whether the candidate has two middle names
- Whether the candidate is left-handed
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How much would you trust a model with dozens of variables like these?

A second yardstick: root mean squared error

The **mean squared error** is the average squared residual:

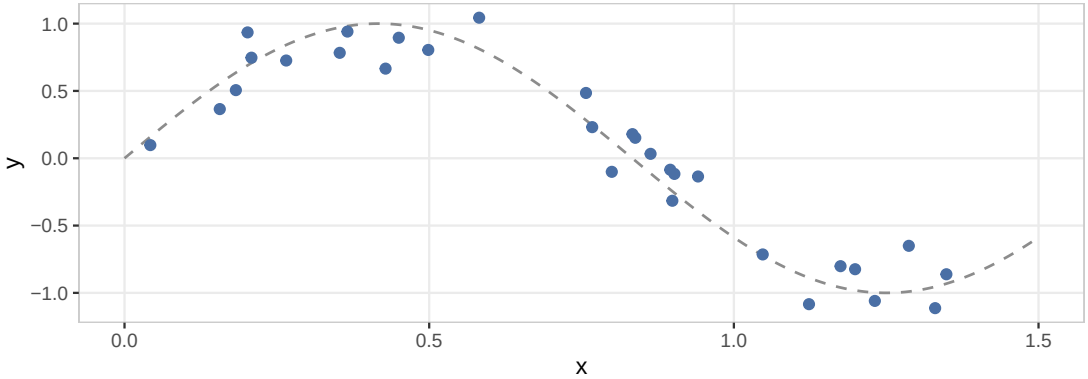
$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2$$

Take the square root and it is back in the units of Y:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}$$

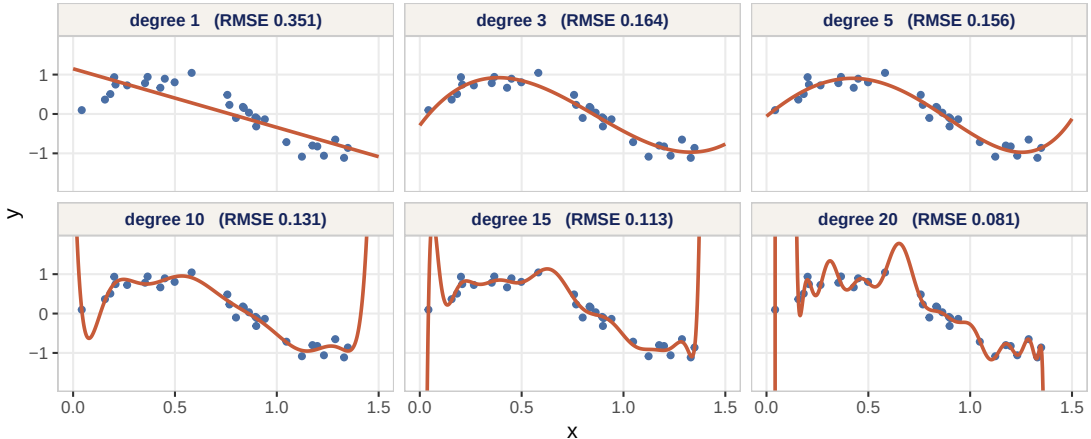
Lower is better: RMSE is the typical size of a prediction error.

A toy problem: a wave plus noise



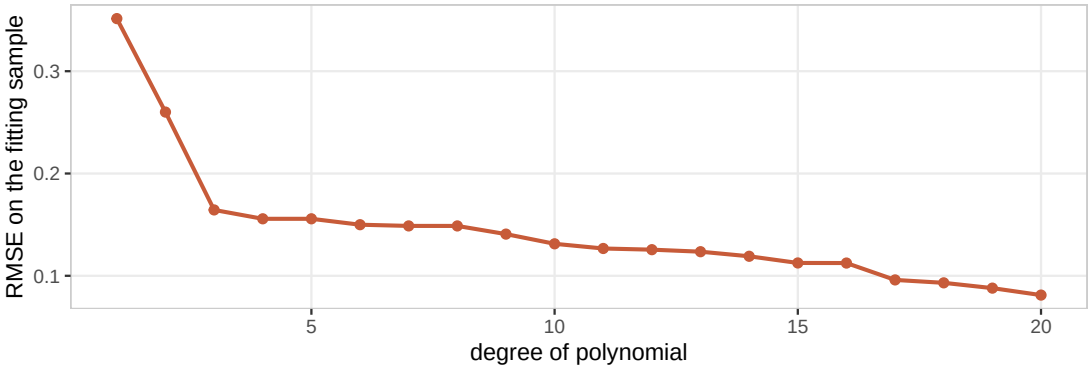
30 points from $y = \sin(1.2\pi x) + \text{noise}$. The dashed curve is the truth we pretend not to know.

Fitting ever more flexible polynomials



As the degree goes up the curve wiggles to chase the sample, and the RMSE keeps falling.

On the data we fit, more is always better

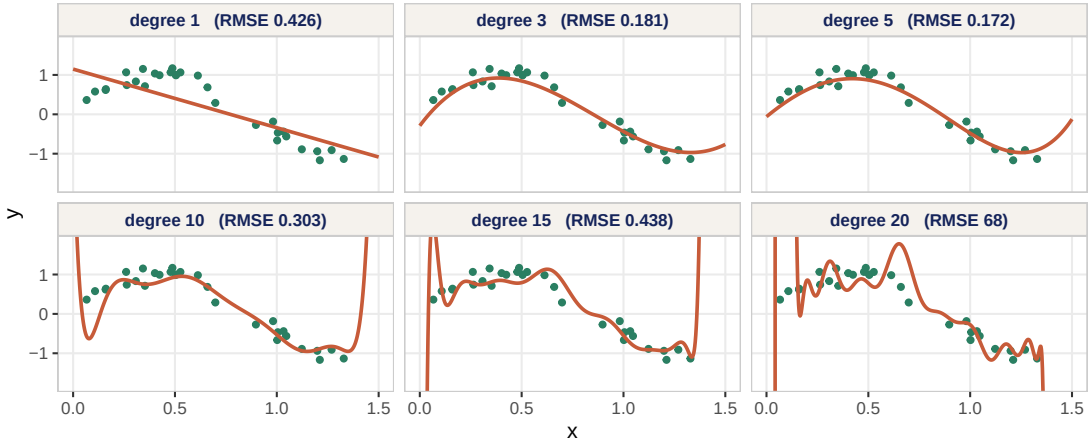


Error on the fitting sample can only go down as the model gets more flexible.

Parsimony

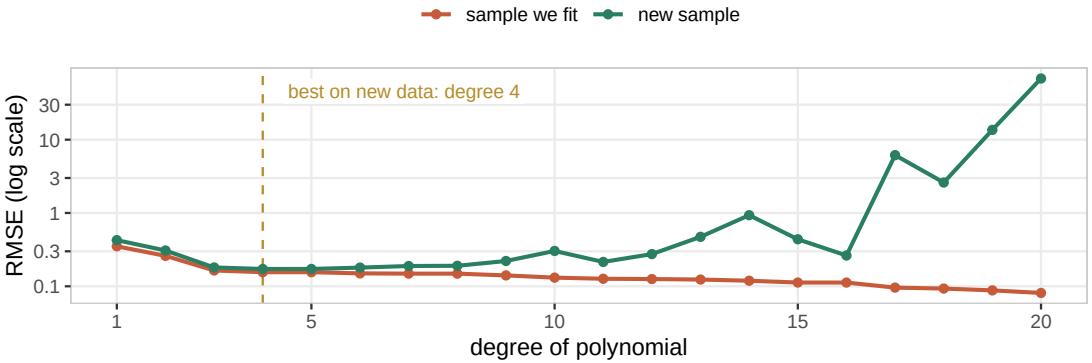
- The best models are **parsimonious**: they balance fitting well against staying simple.
- Simple models are easier to explain, and they tend to generalize better to new data.
- So let's test that. Draw a **fresh sample** from the same process and score the same six curves on it.

The same curves, scored on new data



The curves are unchanged; only the points are new. Now the wiggliest curves pay for their wiggles.

Overfitting, in one picture



Past degree 4, extra flexibility fits the noise in the sample, not the signal, and new-data error climbs.

Overfitting in multiple regression

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- The same thing happens when we choose among many predictors.
- Adding variables always raises R^2 (and lowers RMSE) on the data used to fit the model.
- But a model loaded with variables is more likely to be **overfitting**, and it will not generalize as well.
- A model almost always looks better on the data it was built from than on new data.

To avoid overfitting, use a fit metric that charges a price for every predictor you add.

Adjusted R^2

Adjusted R^2 penalizes R^2 using the number of predictors k and cases n:

$$\text{Adjusted } R^2 = 1 - (1 - R^2)\,\frac{n-1}{n-k-1}$$

- A new predictor must raise R^2 by enough to beat the larger k, or adjusted R^2 falls.
- It keeps roughly the "share of variance explained" reading of R^2.
- Higher is better.

AIC: the Akaike information criterion

$$\text{AIC} = 2k - 2\log(L) \quad \xrightarrow{\text{linear regression}} \quad \text{AIC} = n\log\left(\frac{\text{RSS}}{n}\right) + 2k$$

- k is the number of parameters (predictors plus the intercept), L the likelihood, RSS the residual sum of squares.
- The linear-regression form drops a constant that is the same for every model, so only **differences** in AIC mean anything.
- Lower is better.

Intuition behind AIC

$$\underbrace{n\log(\text{RSS}/n)}_{\textbf{fit}} \quad + \quad \underbrace{2k}_{\textbf{complexity}}$$

fit	**complexity**
falls as predictors are added	rises by 2 per parameter

A predictor earns its place only if it improves the fit by more than the penalty it costs.

There are a *lot* of possible models.

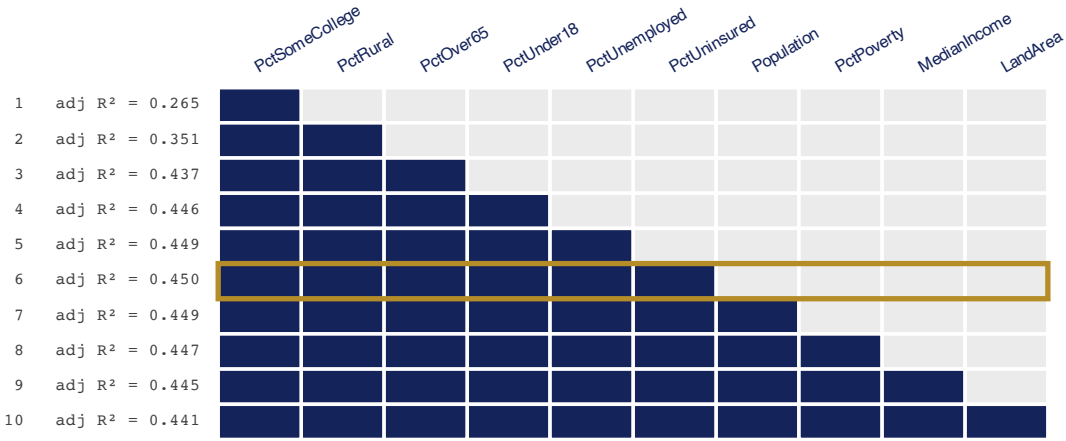
How do we find the one with the highest adjusted R^2 or the lowest AIC?

Best subsets regression

- Computers are fast. Let R fit all $2^k - 1$ models with at least one of the k predictors.
- For each model size, keep the best one; then compare the winners by adjusted R^2 or AIC.

```
library(leaps)
subsets <- regsubsets(PhysiciansPer10000 ~ ., data=counties, nvmax=10)
summary(subsets)$adjr2
```

The winner at every size



Each row is the best model of that size. Adjusted R^2 peaks at 6 predictors (gold), then falls.

The catch: the count explodes

predictors	models to fit
10	1,023
20	1,048,575
40	1,099,511,627,775

Every new predictor **doubles** the work. Past a few dozen, even a fast computer cannot try them all.

Backward stepwise regression

1. Start with the **full** model containing every predictor.
2. Try removing each predictor in turn, and compute the AIC of each smaller model.
3. If any removal lowers the AIC, make the best one and go back to step 2. If none does, stop.

About $k^2/2$ fits instead of 2^k: with 40 predictors, roughly 800 instead of a trillion.

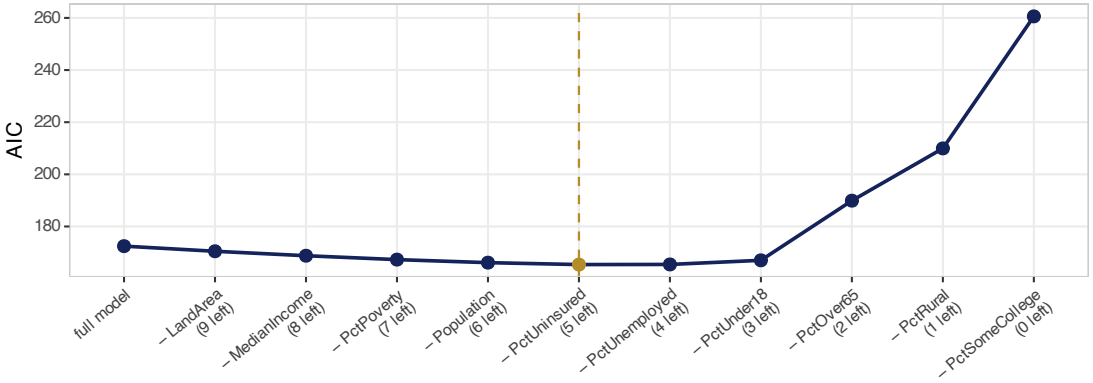
Running it in R

```
full <- lm(PhysiciansPer10000 ~ ., data=counties)
step.model <- step(full, trace=0)
formula(step.model)

PhysiciansPer10000 ~ PctRural + PctUnder18 + PctOver65 + PctSomeCollege +
    PctUnemployed
```

`step()` drops 5 predictors and keeps 5. Set `trace=1` to watch every step.

Watching the elimination



Read left to right: each step removes one predictor. AIC bottoms out at 5 predictors, which is where `step()` stops.

The catch: one step at a time

- Stepwise makes the move that looks best **right now**, and never revisits it.
- It can get locked into a path that never visits the best model.
- Here its 5 predictors are exactly the best 5-predictor subset. That is luck, not a guarantee.

- Notice adjusted R^2 preferred 6 predictors and AIC prefers 5. **Different criteria, different answers**, and both curves are nearly flat from 4 to 6 predictors.

We have been grading our own homework

- So far every metric scores a model on the same data used to fit it.

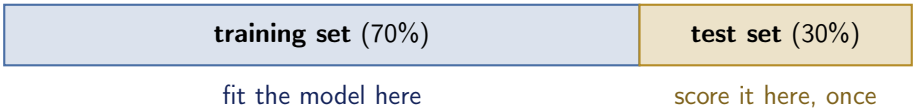
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- So far every metric scores a model on the same data used to fit it.
- That is a little like cheating. It is exactly how the xkcd "rules" get made.
- What we actually care about is how the model does on **new** data.

A solution: hold some data out



- Build every model on the **training set** only.
- Measure RMSE on the **test set**, which the model has never seen.
- That is an honest estimate of how the model will do on new data.

Splitting the counties

```
set.seed(2026)
test.cases   <- sample(1:nrow(counties), 50)  # about 30% of 168
test.set     <- counties[test.cases, ]
training.set <- counties[-test.cases, ]
```

`set.seed()` makes the random split reproducible: same seed, same split.

Full versus stepwise, chosen on the training set

```
full    <- lm(PhysiciansPer10000 ~ ., data=training.set)
reduced <- step(full, trace=0)

rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
rmse(test.set$PhysiciansPer10000, predict(full, test.set))

[1] 1.788

rmse(test.set$PhysiciansPer10000, predict(reduced, test.set))

[1] 1.824
```

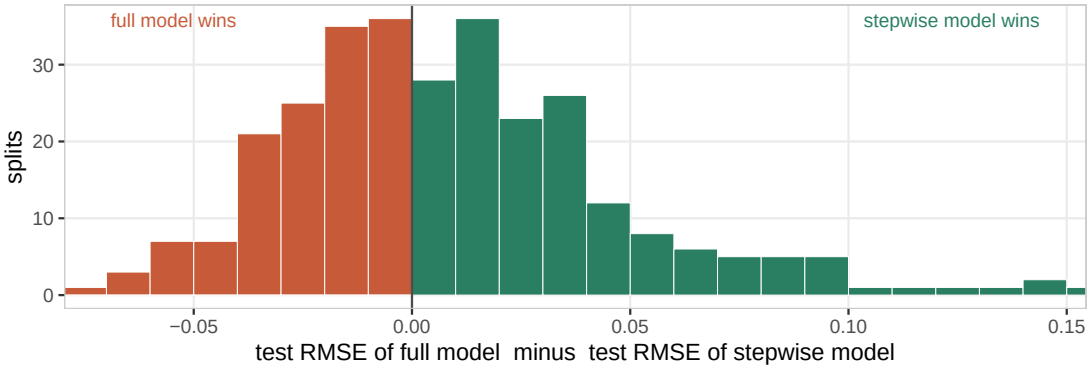
Stepwise runs on the training set too. Choosing variables with the test set would leak it into the model.

The scorecard

	Training RMSE	Test RMSE
Full model (10 predictors)	1.54	1.79
Stepwise model (4 predictors)	1.57	1.82

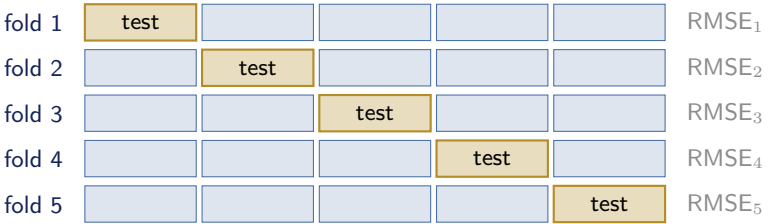
- On the training set the full model wins, as it must.
- On the test set the stepwise model does worse.

But that was one split. Here are 300.



Stepwise wins 54% of the time. One random split can tell almost any story.

Cross-validation: every case gets a turn in the test set



- Cut the data into K folds. Fit on $K-1$ of them, score on the one left out, and rotate.
- Average the K test errors. Every case is predicted exactly once, by a model that never saw it.

The verdict from 10-fold cross-validation

	10-fold CV RMSE
Full model (10 predictors)	1.68
Stepwise model (chosen inside each fold)	1.66

- Essentially a tie. With 168 counties and only 10 predictors, there is **little room to overfit**.
- Give the model more freedom (interactions, many more predictors) and that changes fast. That is where penalized regression comes in.

What to take away

- R^2 and in-sample RMSE always reward complexity. Use **adjusted** R^2 or **AIC** to charge for it.
- **Best subsets** searches everything but does not scale; **stepwise** scales but is greedy.
- The honest test is **new data**: a held-out test set, or better, cross-validation.
- Aim for the parsimonious middle, and let subject knowledge overrule a criterion when it should.